BACHELOR‘S THESIS ASSIGNMENT

I. Personal and study details

Student's name: **Nos Mikahel** Personal ID number: **525799**

Faculty / Institute: **Faculty of Electrical Engineering**

Department / Institute: **Department of Electrical Power Engineering**

Study program: **Electrical Engineering and Computer Science**

II. Bachelor’s thesis details

Bachelor’s thesis title in English:

Design of microcontroller solution for time synchronisation in UAV swarms

Bachelor’s thesis title in Czech:

Návrh mikrokontrolérového řešení pro časovou synchronizaci v rojích dronů

Name and workplace of bachelor’s thesis supervisor:

Ing. Vojtěch Vrba Multi-robot Systems FEE

Name and workplace of second bachelor’s thesis supervisor or consultant:

RNDr. Petr Štěpán, Ph.D. Multi-robot Systems FEE

Date of bachelor’s thesis assignment: **12.02.2026** Deadline for bachelor thesis submission: ____________

Assignment valid until: **19.09.2027**

Zdeněk Müller Digitálně podepsal(a) Zdeněk Müller Datum: 13.02.2026 08:56:23 ID: 307983

Head of department’s signature

Veronika Sobotíková Digitálně podepsal(a) Veronika Sobotíková Datum: 14.02.2026 08:00:11 ID: 31854

Vice-dean´s signature on behalf of the Dean

III. Assignment receipt

The student acknowledges that the bachelor’s thesis is an individual work.
The student must produce his thesis without the assistance of others, with the exception of provided consultations.
Within the bachelor’s thesis, the author must state the names of consultants and include a list of references.

14.02.2026

Date of assignment receipt

Nos Mikahel

Student’s signature

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Guidelines:

Design a printed circuit board, including firmware, that contains the necessary control circuits and a microcontroller for processing time data from GPS RTK modules and potentially other time signal sources (e.g., DCF 77). The goal is to provide a unified time base to various components on all drones in the swarm. Evaluate the functionality and accuracy of the proposed solution in a real experiment with drones.

Bibliography / sources:

1. H. Shim, H. Joo, K. Kim, S. Park and H. Kim, "Provisioning High-Precision Clock Synchronization Between UAVs for Low Latency Networks," in IEEE Access, vol. 12, pp. 190025-190038, 2024, doi: 10.1109/ACCESS.2024.3516856.